* $T(x,y) = (x+y,\ 2x+3y)$

$S : x^2+y^2 = 1$

$(s,t) = (x+y,\ 2x+3y)$

$\Rightarrow s = x+y \Rightarrow 2s = 2x+2y$

$t = 2x+3y$

$\Rightarrow y = t-2s$

$x = s-y = s-t+2s = 3s-t$

$T(s,t) = (3s-t)^2 + (t-2s)^2 = 1$

$\Rightarrow 9s^2 - 6st + t^2 + t^2 - 4st + 4s^2 = 1$

$\Rightarrow 13s^2 - 10st + 2t^2 = 1$

$T^{-1}(S) = (x+y)^2 + (2x+3y)^2 = 1$

$\Rightarrow x^2 + 2xy + y^2 + 4x^2 + 12xy + 9y^2 = 1$

$\Rightarrow 5x^2 + 14xy + 10y^2 = 1.$

$\therefore T^{-1}(S) : 5x^2 + 14xy + 10y^2 = 1$

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